

MyDesign Portfolio

Element D Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design ideas, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design ideas, be sure to delete the instructions.

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to

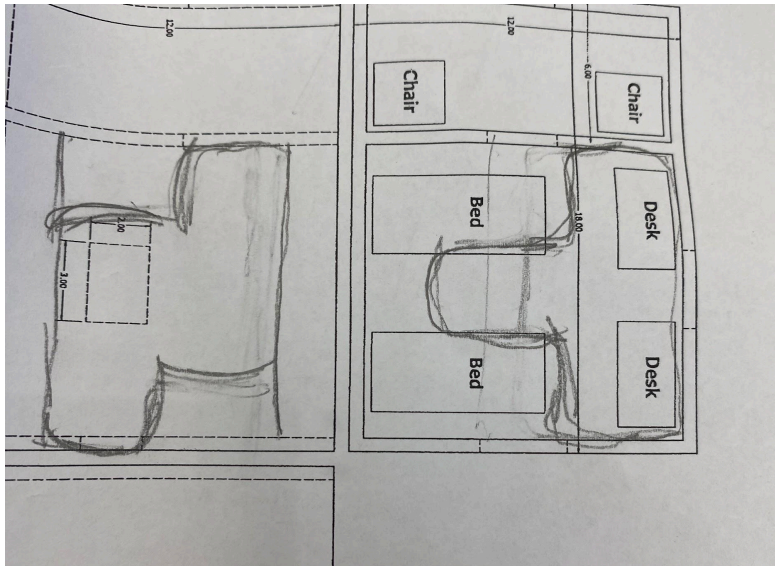
viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Design Idea #1: sand balloon

Date: 03.18.26

Description: a rubber balloon full of sand that when the heat flows up the balloon will pop and the fire will be distinguished by the sand.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

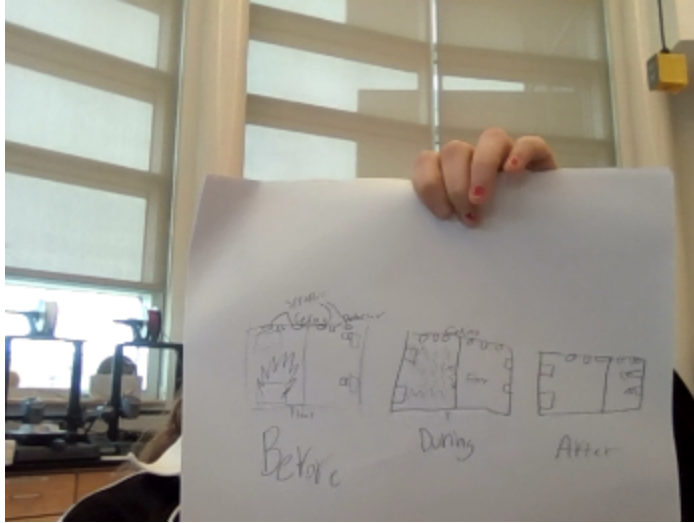


Design Idea #2: sprinkler

Date: 03.18.26

Description: you have a smoke detector that once the smoke is detected the sprinklers activate and the water sprays down and eliminates the fire. There is a detector and a set of sprinklers in each room.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

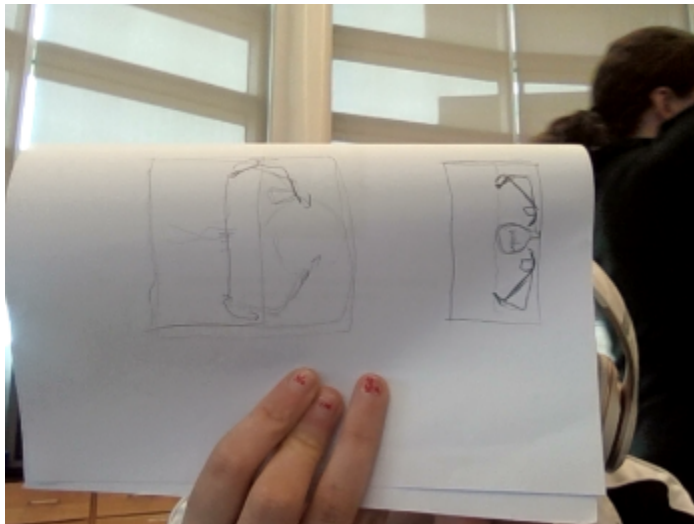


Design Idea #3: water drop

Date: 03.18.26

Description: A balloon filled with water that is skewered with kabob sticks that are attached to a string. Once the fire grows tall enough to cut the string, the kabob sticks will pop the balloon and extinguish the fire.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection (D1, D2)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it

allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

	Weight 1-10	Testing setup and procedure	Number of trials	Expected results	pass/fail criteria, goal
Criterion A→ The fire is completely extinguished, with no flame visible.	10	Place a lit candle inside the apartment. Check for flame after the suppression system is finished.	# trials: 2	The fire will be completely killed and will not ignite again after extinguishment.	Pass: The fire will be completely killed.
Criterion B→The fire is completely extinguished within 5 minutes.	10	Place a lit candle inside the apartment. Time how long the suppression system takes to kill the fire.	# trials: 2	The fire will be subdued quickly.	Pass: The fire will be subdued within 5 minutes.
Criterion C→The detection system detects the fire within 30 seconds.	8	Place a lit candle inside the apartment. Time how long it takes for the lights on the thermosensor to go off.	# trials: 2	The alert system will start sounding off and the lights will start flickering once the system detects the fire quickly.	Pass: The alert system goes off in 30 seconds.
Criterion D→ The alert system goes off within a 30 second window.	8	Place a lit candle inside the apartment. Time how	# trials: 2	The alert system will sound off and the lights will flicker once	Pass: The alert system goes off once fire is detected.

		long it takes for the lights and buzzer to go off.		the fire is detected.	
Criterion E→ Fire remains in one room.	10	Place a lit candle inside the apartment. Observe the spread of the fire.	# trials: 2	The fire is detected and extinguished fast enough, so the fire does not destroy a lot of furniture.	Pass: The fire is contained in one area.
Criterion F→ The alert system does not go off by anything other than the light emitted from the fire.	7	Observe that the thermosensor turns its lights on when bright lights are near it.	# trials: 2	The alert system is set off by the fire alone.	Pass: The alert system is set off by the fire alone.
score	53				

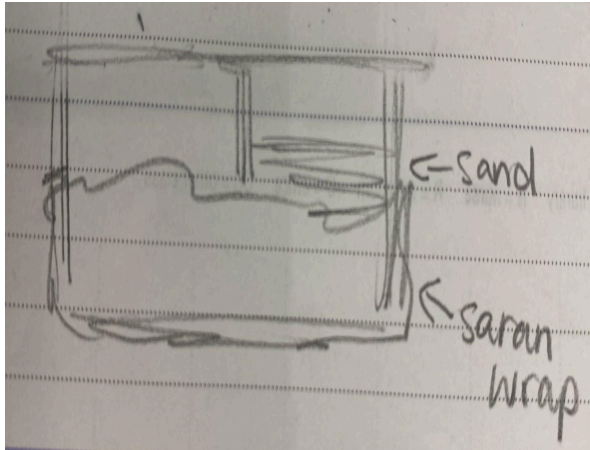
Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: 3/18/2026

Sketch:



Manufacturing Methods:

Step 1: 3d print a frame with a grid pattern on the inside.

Step 2: Wrap one side with saran wrap.

Step 3: Fill the inside with sand.

Supply List and Costs:

Item	Cost each	# needed	Total
Party animal balloon	\$0.05	2	\$0.10
Sand	\$0.15/lb	.5 lb	\$0.08
Arduino	\$27	1	\$27
Total Cost			\$27.18

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

 Group 1: Gantt chart